

Subtle Green Grabbing? The Extractive Dimensions of Carbon Offsetting

by
Andrea Rizzi

Among the subtlest forms of contemporary “green grabbing” are carbon offsetting ventures, as most nature-based carbon credits are issued in tropical countries through carbon forestry projects. While it may appear less problematic than the seizing of tangible land, the “scramble for carbon” unleashed by global “green” agendas implies new de facto enclosures—often on Indigenous land—and requires the disciplining of forest communities. Drawing on research in the Colombian Amazon, this paper brings into relief the territorial and extractive dimensions of carbon offsetting, as well as the forms of resistance it generates.

Entre las formas más sutiles del “acaparamiento verde” contemporáneo destacan los negocios de compensación de emisiones del carbono, ya que la mayoría de los bonos del carbono basados en la naturaleza se producen en países tropicales mediante proyectos de reforestación. Aunque a primera vista puedan parecer menos problemático que la apropiación tangible de tierras, la “carrera por el carbono” desatada por las agendas climáticas globales conlleva nuevos cercamientos de facto —a menudo en tierras indígenas— y requiere disciplinar las comunidades rurales. Basándose en investigaciones realizadas en la Amazonía colombiana, este artículo pone de relieve las dimensiones territoriales y extractivas de los proyectos de compensación de emisiones de carbono, así como las formas de resistencia que generan.

Keywords: Carbon offsetting, Extractivism, Green grabbing, Indigenous communities, Colombia

When we think of land grabbing, we tend to think of an illegal, para-legal, or violent acquisition of land for extractive purposes that are typically environmentally destructive. Over 10 years ago, the framework of “green grabbing” (the appropriation of land for environmental ends) called attention to an emerging process of land appropriation, mostly in the Global South, on the back of UN-sanctioned environmental agendas (Fairhead, Leache, and Scoones, 2012). The risk, Fairhead and colleagues warned, was one of “colonial and neocolonial resource alienation in the name of the environment” (2012: 237) through schemes ostensibly aimed at biodiversity conservation, biocarbon sequestration, biofuels, ecosystem services, ecotourism, or offsetting. Arguably, the subtlest among these is carbon offsetting.

In the existing scholarship, carbon offsetting projects often appear either as “black boxes” in studies taking a global perspective on carbon markets or, conversely, as if they took place in a vacuum in case study analyses (Rizzi, 2025).

Andrea Rizzi is a PhD candidate at the University of Bologna (geography cluster).

As a result, it takes little effort for practitioners to create a “self-referential space where members of epistemic communities cite and reinforce each other’s interpretation” (Büscher, 2014: 87). As Moros, Corbera, Vélez, and Flechas (2020) point out regarding payments for ecosystem service (PES), with an argument that could apply to the green grabbing literature, academics endorsing a critical discourse says more about their acquaintance with relevant conceptual debates than about the existence of “hard” evidence supporting that discourse.

Heeding Fairhead, Leach, and Scoones’ advise not to lose sight of the “empirical particularities of diverse cases” (2021: 241), this paper aims to make some of these connections explicit by showing how mitigation capital is put to work in the Colombian REDD+ landscape, and by bringing into relief the contradictory ways in which it is embraced and negotiated by Indigenous communities. Drawing on existing literature in geography, political ecology, and cognate disciplines, as well as on field research in Colombia, the paper argues that, far from being a purely abstract financial phenomenon, nature-based carbon offsetting presents an unneglectable material impact that, through a biased identification of deforestation drivers, can amount to a subtle form of green grabbing. In the tradition of political ecology, this paper contributes to dispelling some of the fog surrounding the situated effects of carbon forestry and the making of environmental subjects upon carbon extractive frontiers, calling into question the historical, environmental, and epistemic justice of the carbon trading paradigm.

The paper is structured as follows: after a brief methodological section, I introduce accumulation by decarbonization and the neocolonial traits of the carbon offsetting regime, and demonstrate its relevance to the green grabbing debate. Next, I make a case for considering the materiality of nature-based carbon offsets, as well as the historical processes through which this commodity was made. I then move on to sketch out the carbon offsetting landscape in Colombia, and address both the material and the immaterial impact of carbon projects through several concrete examples. The paper ends with concluding reflections.

METHODOLOGY

This paper is based on doctoral research in which the data and reflections stem from the following sources: a review of relevant literature in the fields of geography, political ecology, development studies, and cognate disciplines; an analysis of relevant documents made publicly available by project developers, government agencies, NGOs, etc., as well as unpublished documents acquired by the author; and field research in the Guaviare department, including over 40 qualitative interviews with institutions, NGOs, and practitioners, group discussions with community members, and structured observation. While this research focused on one specific case study, this paper takes a broader perspective on Colombia and does not make an extensive use of the primary material, as the paper was written as fieldwork was still in progress.

As discussed below, the lack of transparency in the carbon offsetting industry and—specifically for Colombia—the absence of a functioning official

registry constitutes a considerable hinderance for social science research. This forces scholars to rely on piecemeal data, semi-official reports (for instance, by NGOs or media), and individual case-studies. In this sense, the role of research is precisely to make sense of this archipelago of bits of data to attempt and draw a comprehensive picture.

This lack of transparency is compounded by regulatory differences among departments and countries, which makes it difficult to offer a region-wide overview. For this reason, as well as not to overextend the scope, this paper does not venture into a comparative analysis of carbon offsetting in different Latin American countries. However, those who are familiar with carbon forestry in other contexts will be able to appreciate similarities and differences against the backdrop of the Colombian forest carbon landscape.

NEOCOLONIAL ACCUMULATION BY DECARBONISATION

Legitimized by the “decarbonization consensus” (Bringel and Svampa, 2023), governments, international organizations, and private actors have been setting up legal infrastructures and funding mechanisms for the trade of carbon credits. This response to the climate problem, termed by scholars “accumulation by decarbonization,” is openly neoliberal in that it “devolve[s] governance over the atmosphere to supranational and nonstate actors and to the market” (Bumpus and Liverman, 2008: 127). In the carbon economy, this accumulation takes up neocolonial traits, as a large portion of carbon offsets is issued through carbon forestry in the Global South (Dufrasne, 2023). This is not by chance. As Gutiérrez (2011: 654) points out, a “market in emission reductions can only be conceived in the context of uneven development,” a context in which offsetting emissions is cheaper in certain regions of the world as it is premised (and maps onto) existing inequalities. As scholars have demonstrated, however, the global divide in the cost of labor power is by no means natural, but in fact historically produced (see particularly Amin, 1976).

The carbon offsetting paradigm is in perfect consonance with the “economy of repair” evoked by Fairhead, Leach, and Scoones (2012: 242), in which the “damage inflicted by economic growth generating unsustainable resource thus creates the basis for the new growth economy of repair.” Leonardi (2017: 79) goes further along these lines by affirming that “carbon markets, in order to be offered as their own remedy, must always fail to a certain extent.” This is but a testament to the ability of capital to take advantage of the recurring systemic crises by reinventing itself in ways that spawn new crises to be exploited.

The “scramble for carbon” described above has brought to the fore the issue of carbon tenure or, simply put, what and whose CO₂ is being valorized. While in some countries governments may seek to ensure centralized control over the carbon held in national ecosystems (Phelps, Webb, and Agrawal, 2010),¹ in countries with a more robustly decentralized (or contested) forest tenure landscape, as is the case in most of Latin America, carbon industry actors have to negotiate access to carbon stocks at the smaller scale of rural localities. Though it may appear less problematic than the seizing of tangible land, the appropriation of carbon stocks may constitute not only an “invisible” land grab, but also

a “Trojan horse” for actual land grabs through the establishment of new de facto enclosures and the disciplining of forest communities. This is particularly true in the case of REDD+ (Reducing Emissions from Deforestation and Forest Degradation), an offsetting method that, despite bad press and critical scholarship (Asiyanbi and Lund, 2020), remains the most widespread among nature-based offsetting methods. These methods, also called forest and land use methods, accounted for approximately 46 percent of traded carbon credits and for 64 percent of traded carbon value in 2022 (MSCI, 2023).

THE MATERIALITY OF CARBON OFFSETS

The academic literature tends to emphasize the abstract financial dimension of carbon credits (Leonardi, 2017) and their virtual essence (Cavanagh and Benjaminsen, 2014). In fact, unlike “technology-based” credits (renewable energies, waste management, etc.), nature-based carbon credits display an undeniable material dimension which, in many respects, resembles that of “traditional” tangible resources. Indeed, nature-based carbon credits do not represent an imaginary unit of carbon that would have been emitted, for example, by burning fossil fuels in a hypothetical future, but rather one that can be pinned down to a relatively bounded ecosystem. This representational and physical act of separation, which Castree (2003) terms “individuation,” suggests that not all carbon commodities are the same. Focusing only on the abstract essence of credits as fictitious capital, means neglecting their material “precursors” (a tree, a tract of peatland, etc.), and, more importantly, leaving their effects on the ground unattended. These include the long-term physical effort required to “extract” credits (i.e. conservation activities, mostly entrusted to local communities as project “owners”) and the territorial “disciplining” processes this entails, which typically take the form of restrictions to forest access and use.

The “physical” dimension of the carbon commodity is surely not lost on governments throughout the Global South, which are moving to claim control over carbon stocks as strategic national assets even in areas where land tenure is formally devolved to Indigenous communities, and at times are striking large-scale deals with global private carbon actors (Marawanyika and Sguazzin, 2023; Kavanagh, 2023). Indigenous communities themselves often perceive the inescapable materiality of these new, ephemeral commodity frontiers as carbon is “materialized” for them by intermediaries: “What kind of pipes or other things do we need to collect the carbon? . . . do we need to collect it in plastic bags?” asks a Papua New Guinean landowner cited by Leggett and Lovell (2011: 11). Both they and Setyowati, in a paper published almost 10 years later, note that Indonesian villagers believe they are “selling the wind” (Setyowati, 2020: 466).

Tending to the physical effects of commodified carbon is not a scholarly whim. Only by understanding carbon as a commodity that is “extracted” and “exported” from areas that are particularly rich in this resource is it possible to conceive the full scale of the above-mentioned “accumulation by decarbonisation” (Bumpus and Liverman, 2008) and to identify the ways in which capital is opening up new resource frontiers as it is pushed away, at least in part, from

“old” ones (fossil fuel extractive frontiers). Put differently, the decarbonization consensus buttressing the reproduction of the capitalist mode of production at a time of climate crisis is predicated upon purported solutions such as carbon offsetting schemes.

To those studying or living in extractive frontiers, this phenomenon looks well-rehearsed: the regions that, based on pre-existing socio-natural conditions, are best endowed with the resource of the day. In this case, the socio-natural environments conceived of as forests (Gutierrez, 2011: 646), which Indigenous communities have not only stewarded but also cultivated over the centuries,² are construed as “rich in carbon (offsets).” It would be wrong, however, to believe that this is only grounded in objective, natural criteria, such as the high carbon uptake of rainforests. At least three other criteria co-constitute the carbon commodity historical polluters are after: first, a relatively low population density, which reduces urbanization pressure on ecosystems and allows large areas to be allocated to carbon storage; second, low opportunity-costs for forest communities to switch or cede their forest uses, which, especially at times of low carbon prices, guarantees the economic viability of carbon projects; third, the so-called “Project Design Document” that, through highly technical calculations and projections, conjures up the prospect of future deforestation, thereby materializing the compensation. This is presumably what Bridge had in mind when he observed that resources do not just “exist,” but rather are made when they “simultaneously ‘fit’ existing socio-technical arrangements and solve certain problems” (2009: 1220).

Paradoxically, in rainforest countries the new (carbon forestry) frontier often coexists with an apparently incompatible resource frontier, namely the deforestation frontier, typically for timber, agrobusiness, and illicit cultivations. Yet this is the natural “habitat” of carbon forestry, whose *raison d’être* are threatened ecosystems. Let us now see how these processes unfold in the Colombian Amazon.

CARBON OFFSETTING IN COLOMBIA

This section offers an overview of the carbon offsetting landscape in Colombia. Before analyzing the carbon industry, the related issues of land and deforestation in the country are briefly outlined.

Land in Colombia has always been bitterly contested. While the land issue cannot be comprehensively discussed here for reasons of space,³ for the purposes of this paper some key aspects are worth mentioning. Land tenure in Colombia has long been insecure due to conflict and, in general terms, land grabbing (Thomson, 2019), even though the land rights of Indigenous and Afro-descendant communities are enshrined in the 1991 constitution (Velásquez Ruiz, 2018). Today, Colombia hosts over 800 *resguardos*, territorial units collectively owned by an Indigenous community (Mendoza, 2024). This, however, is only on paper, as illegal actors and the expansion of extractive frontiers make the *de facto* reality rather different; regardless of this discrepancy, formal land tenure plays a key role in forest-based carbon offsetting.

Forests cover around half of the Colombian territory, about two thirds of which is in the Amazon region. The Amazon also accounts for most of the deforestation in the country (at least 70 percent according to Rodríguez de Francisco et al., 2021). Although 2022 and 2023 witnessed a widely celebrated improvement (Ministerio de Ambiente y Desarrollo Sostenible, 2024), it consisted basically of a reduction in deforestation rates, not to an increase in forest cover.

Within this context, some have come to see carbon forestry as a win-win solution for both the country's forests and some of its most vulnerable and impoverished populations. Unlike other countries in the region, which have either rejected carbon commodification or sought to develop jurisdictional markets for a regulated sale of offsets,⁴ Colombia has been taken by storm by carbon offsetting. This started when the government led by Juan Manuel Santos established a carbon tax and almost simultaneously offered the opportunity for businesses to obtain tax breaks through offsetting (Dufrasne, 2021), thereby generating a national carbon market. President Santos, however, as well as his successors Iván Duque and Gustavo Petro, have failed to regulate this market. Colombia does not even have a reliable official registry of active carbon projects, yet unofficial accounts reveal an explosion in projects for the voluntary carbon market. In early 2025, the "Renare" registry (see below) included 195 projects (Renare, 2025). Unofficial estimates vary considerably. According to the Colombian carbon industry association, in 2023 the country had 208 certified projects (80 reforestation plus 47 REDD+, see Asocarbo, 2023). A report by Colombian NGOs speaks of 123 ongoing projects in the country, of which 52 are afforestation/reforestation and 71 are REDD+ (Pérez Lara et al., 2023), while the "Geo-Grafiar" portal, launched in 2024, features 92 active REDD+ projects country-wide (CENSAT, 2025). According to Instituto Sinchi, the Colombian Amazon alone hosted 51 projects in 2022 (Díaz and Ruiz-Nieto, 2023).

This is no minor detail: in 2016, the Colombian Ministry of the Environment and Sustainable Development launched "Visión Amazonía," a jurisdictional forest conservation program covering the whole Colombian Amazon biome. A REDD Early Movers (REM) program supported by Germany, Norway, the United Kingdom, and the Green Climate Fund, its aim is to manage REDD+ in line with the provisions of the United Nations Framework Convention on Climate Change (UNFCCC). Unlike voluntary offsetting schemes, jurisdictional programs require all offsetting projects to be "nested" under the program and to determine their baselines in compliance with the government's Forest Reference Emissions Level (FREL), a document establishing the expected rate of deforestation. A comprehensive *Registro Nacional de Reducción de las Emisiones de Gas de Efecto Invernadero* (National Registry of Greenhouse Gas Emissions Reduction, Renare) was supposed to avoid double counting and therefore guarantee that international donors were effectively contributing to additional mitigation projects.⁵ This sparked controversy in 2015, when the large Matavén project claimed to have reduced emissions, on a per hectare basis, by nine times more than the jurisdictional project (Dufrasne, 2021). After Renare was taken down amid criticism by civil society organizations and market actors, a "2.0" version was launched in May 2024. It is unclear, however, in what ways the

new version is an improvement to the original one, as the information listed is provided by carbon project operators on a self-certification basis. This means that no official authority monitors existing projects, their implementation, nor their potential overlaps (with each other or with “Visión Amazonía”).

This appears particularly worrying when considering that Colombia hosts 6-8 percent of the Amazon ecosystem⁶ and approximately 100 different Indigenous peoples (ONIC, n.d.). Most projects claim an involvement of Indigenous groups, since the planning stages state the empowerment of these communities among their goals. Investigative reports (Garzón, Galvis, and Díaz, 2023), and a shaky track record in terms of inclusionary/exclusionary dynamics characterizing processes of environmental change in the country (Ulloa, 2013), however, have cast doubts on the extent to which these ventures can really benefit forest communities.

THE MATERIAL IMPACT OF CARBON OFFSETTING

This section teases out the material impact of nature-based carbon offsetting in Colombia. It does so by presenting several concrete cases and thoroughly examining one case that was investigated on the ground.

In Colombia, most carbon projects are not the result of bottom-up initiatives by local communities, but rather are implemented in reaction to ready-made offers by external actors to the communities that hold collective property rights over the territory being marketed. Scholars and civil society organizations have repeatedly called into question the authenticity of Indigenous involvement in designing, steering, and sharing the benefits of such projects (Cardona, 2014; Ulloa, 2013).

Another aspect of the carbon business that is often underemphasized is that, as (Tienhaara, 2012: 557) remarks, “forest sequestration projects, particularly those employing avoided deforestation, are much less capital-intensive than extractive projects (e.g. mining, oil and gas).” This means that speculation in carbon is open even to minor “soldiers of fortune” among capitalists, also called “carbon cowboys.” On a deeper level, the low barriers to entry that characterize the carbon business, in Colombia and elsewhere, are conducive to new forms of internal colonialism (Casanova, 1965). Within the carbon economy, this takes the shape of urban capitalists buying up rights to pollute from rural communities (Bruna, 2022). To add insult to injury, the emission reductions might never actually take place (or not be additional, which for mitigation purposes makes little difference), yet Indigenous livelihoods are altered and most of the value extracted from the socio-ecological process underpinning credit issuance is captured along the opaque carbon commodity chain (Dufrasne, 2023).

Arguably, such processes are at play in the Colombian carbon economy. As is well known, Indigenous communities—in Latin America and elsewhere—hold a very broad conception of what modern Western civilization understands as territory, nature, and livelihood. While it is beyond the scope of this paper to analyze these distinctions,⁷ it is a fair approximation to affirm that Indigenous peoples typically perceive themselves as enmeshed in a web of human and

more-than-human relationships, involving spiritual connections that go beyond the biological living/non-living divide (Schroeder and González, 2019). Yet, in carbon commodity frontiers, the multifarious significance of entire ecosystems is singularized for carbon storage. Besides the obvious epistemic implications, this does not diminish but rather reinforces the relevance of land tenure contestations, while adding a layer of carbon tenure, as the two do not necessarily coincide. To many Indigenous communities, as was illustrated above, dealing in carbon equates to dealing in air or wind. Colombian communities are no exception. As research by Gutiérrez, De La Hoz, Hernández, and Silva (2023) has revealed, the Piaroa involved in the large-scale Matavén project in eastern Colombia initially believed the project was about selling air.

Even when land and carbon ownership is settled on paper, the question of representation arises, that is, who has the right to sign a deal in the name of an Indigenous community. This is the bone of contention in the infamous case of the REDD+ Baka Rokarire project. In March 2021, a person claiming to represent the Association of Indigenous Traditional Authorities of the Pirá-Paraná River (ACAPI) in the Vaupés Department signed an agreement with an environmental NGO from the Department of Antioquia. Later that year, Indigenous authorities were taken aback when they found out that “their” carbon was on sale on carbon markets. They took the case to court seeking the contract’s annulment, claiming that the signatory had left his position two weeks before signing. The case reached the Constitutional Court, which could establish a landmark precedent for the development of carbon forestry projects (Bermúdez Liévano, 2022).⁸

Regardless of the court ruling, this case, which is not an isolated one, reveals another complication in the development of forest carbon projects. Who can speak on behalf of Indigenous communities, especially those that are spread throughout the Amazon and traditionally adopt horizontal decision-making methods? Moreover, what forms of regulation and oversight should carbon forestry agreements be subject to, since they constitute a legal contract between a private actor and a formal territorial entity, and not merely among private entities? In another case, concerning the Banakale Isimali REDD+ project, investigations have revealed that the developer has not carried out a prior consultation process as prescribed by the Colombian law (Mayorga Alejo, 2023).

Once all preparatory steps for a project have been fulfilled, the actual offsetting, i.e. the activities supposed to prevent deforestation and/or forest degradation, may begin. For a project to be at least partially effective, though, it must address real deforestation drivers, which then have to be monitored over the course of the entire project (typically several decades). What exactly is considered a source of deforestation, or even simply forest degradation,⁹ is another source of debate.

The Western conservation discourse has long blamed local forest-dependent communities, pointing to minor extractive activities such as hunting, wood harvesting, or shifting agriculture (Skutsch and Turnhout, 2020; Ingalls and Dwyer, 2016). This is often a misrepresentation of the reality on the ground, one in which large-scale deforestation is driven by timber extraction, cattle farming, industrial agriculture, and illicit drug cultivation (Dávalos et al., 2011). While local individuals might be involved in such activities, it is not the livelihood

strategies of impoverished local communities that are to blame, but rather the industrial forest uses that have largely replaced those very subsistence strategies, sometimes generating waves of dispossession that, in turn, forced local inhabitants to turn into industrial laborers. It is in this discursive reversal of the elemental factors of deforestation, “need and greed” (Sponsel, Headland, and Bailley, 1996), that the true essence of the subtle green grabbing taking place through carbon offsetting really lies, as forest communities preserve formal land rights, yet their forest uses are impacted and restricted.

This contradiction has been observed in Colombia, as researchers from Instituto Sinchi report: “One aspect of special interest and concern . . . is the commitment to reduce the area of *chagras* by 30 percent under the zero-deforestation program” (Díaz and Ruiz-Nieto, 2023). The *chagra* is an ancestral agroforestry system adapted to the characteristics of the Amazon environment, which, being the seminal method for the material reproduction of many Indigenous communities, possesses a remarkable cultural and biological value. The former is valued for the “transmission of traditional knowledge about management, sustainable use, and conservation of biodiversity, and . . . about family and social relations [and] the handling of food, ritual, and medicinal plants” (Garavito et al., 2021: 2). The latter is linked to its ability to “strengthen the recovery of degraded ecosystems. . . Far from degrading the soil, the *chagra* produces more fertile soil” (Garzón, Garzón, and Díaz, 2023). Indeed, agroecologists and environmental anthropologists, among others, have long argued that “swidden cultivation is . . . more productive than commercial logging in terms of the size of the population supported” (Dove, 1983: 85).

The same inconsistent logic applies to nomadic Indigenous groups. One particularly telling example is that of the Deiyiabena REDD+ Nükak project in the Guaviare Department (Amazon Carbon, n.d.), which I will turn to next.

DEIYIABENA REDD+ NÜKAK

Guaviare, one of Colombia’s 10 Amazonian departments, hosts a considerable portion of the forest carbon projects currently existing in the country at different stages of implementation. Guaviare is also one of the deforestation hotspots in the region (Global Forest Watch, n.d.), and has historically been one of the departments with the most widespread territorial presence of armed groups—two phenomena which, as Rodríguez de Francisco et al. (2021) and others showed, interlock in nefarious ways for the region’s forest cover, particularly since the 2016 peace agreement. All these circumstances clearly raise questions as to the feasibility, measurability, and permanence of carbon reductions.

Within this context, the Nükak Indigenous group has been identified as a key interlocutor for carbon offsetting. The Nükak, a small (approximately 1,000 members) semi-nomadic community in the so-called state of “initial contact” with Colombian urbanized society (Mahecha and Franky, 2013), traditionally roamed across a territory extending between the Guaviare and Inírida rivers. This territory, approximately 1 million hectares southeast of the departmental capital San José, is designated a *resguardo*.

In the early 2000s, however, they were violently driven out of the territory as the result of processes that lie at the intersection of armed conflict, drug

cultivation, and illicit expansion of the agricultural frontier (Franky, 2011). Since then, they have only been able to access their territory sporadically. This, compounded with their encounter with Colombian “urbanized” society, has cast the Nükak in a dire situation that local institutions and NGOs have either deliberately neglected or were not been able to solve. Today, most Nükak live in the outskirts of San José or near peasant settlements in precarious socioeconomic and health conditions. One social worker begrudgingly stated: “There are so many interests behind the Nükak that nobody is interested in getting them out of their condition. The poorer they are, the more wretched they are, the more they starve, the more aid will come their way, and the more aid local authorities and NGOs will put their hands on” (personal communication, May 2024).

Since 2019, the Nükak have been approached by different carbon project developers seeking to establish a REDD+ project on their land. The first project collapsed under societal pressure due to its openly vexatious conditions, particularly in relation to the circumstances in which the deal was signed and the secrecy demanded of the Indigenous signatories (Mongabay, 2022). The second, started in 2021, fell through due to undisclosed reasons while the author was in the field (one person involved alleged that the developer had “run out of money”). A third attempt, with a different developer and certifier, is currently in the works, and might involve afforestation/reforestation in addition to REDD+.

On paper, the main goal is to preserve the vast forested area of the *resguardo* (almost 800,000 hectares) in partnership with the Nükak by promoting “sustainable forest uses” among community members (Amazon Carbon. n.d.). Here, a double paradox emerges. First, not only are they not a substantial threat to the forest, but they are not in a position to halt deforestation, even though they formally own the land. The institutional and non-institutional (NGO) actors interviewed agreed on the fact that the Nükak’s forest uses are in no way a major deforestation driver, and that the deforestation within their *resguardo* (which peaked between 2016 and 2022) is caused by land grabbing for cattle ranching and, to a lesser extent, timber and illicit cultivations.

Second, even if the Nükak were to regain territorial control over their land, the carbon forestry deal would prevent them from using it freely in their customary ways, as they would be expected to comply with Western sustainability dogmas. Despite these blatant flaws, project preparations have been ongoing as the Nükak have been promised (and in small part given) monetary compensation for their participation, which to some of them seems more than institutions and NGOs are ready to do. In the absence of strict regulations and safeguards for free prior and informed consent, it is ultimately a private negotiation between a developer and community that has the right to express a decision through its leaders regardless of how well they have been informed. As Ulloa (2013) argues, the symbolic and de facto appropriation of territory and nature (effectively, new enclosures) is one of the processes unleashed by the consolidation of what is referred to as “neoliberal natures.”

Scientific research and grey literature seem to indicate that the Nükak REDD+ project in Guaviare is not an isolated case. Indeed, despite all evidence, shifting cultivation and nomadic practices are being demonized in Colombia and worldwide (Asia Indigenous Peoples Pact, 2010). This misconception is useful for

carbon project developers who see Indigenous livelihoods as a low-hanging fruit for conservation deals. In most tropical countries large-scale deforestation is not only an unassailable “Moloch” for developers (being influenced to an appreciable extent only by wider socio-political transitions, see Rodríguez-de-Francisco et al., 2021), but it is at the same time vital to the very existence of carbon offsetting. Indeed, the carbon industry has a perverse incentive not only to overstate, but to allow deforestation to a certain extent, for example in the so-called project reference region (Haya et al., 2023), to be able to vouch for its contribution. What is ironic here is that ancestral forest management methods that have been successfully being used for millennia (Levis et al., 2017) are now blamed for forest degradation by actors who, in the eyes of forest communities, might rightfully be seen as forest “newcomers.”

This section has focused on the controversial establishment of carbon projects in the Colombian Amazon and on its material impact. That, however, is only half the story. Let us now look at the potential immaterial impacts of carbon offsetting on forest communities.

THE IMMATERIAL IMPACT OF CARBON OFFSETTING

After delving into the more material aspects of nature-based carbon offsetting, this section turns to the immaterial, cultural, and gendered repercussions of carbon forestry, zooming out again to the whole of Colombia and Latin America.

In a country that, besides being a former colony, is still paying a very high social toll for the civil war that ravaged it from 1964 to 2016, Indigenous communities are among the most impoverished, vulnerable groups. Though with remarkable differences across the country, the challenges faced by Indigenous communities include poverty, addictions, health issues, and cultural vulnerability. While on paper carbon forestry programs address all of these, their development can affect forest communities in problematic ways, from entrenching existing power imbalances to impinging both on their material subsistence and social reproduction.

Officially, REDD+ projects are being developed through participatory processes. One might wonder, nonetheless, how participatory any such project can be in dealing with “ethnic” groups composed by communities that are sometimes separated by hundreds of kilometers of dirt roads or boat trips, whose members may have limited understanding of Spanish, and no understanding about the financialization of nature underpinning carbon markets. According to an environmental-NGO worker: “I do not even think it is clear to the Indigenous leaders who are negotiating, but the other Indigenous people in the forest, ordinary families, do not have the slightest idea. Their world is damaged by the white man [sic], by cows, by coca, by the things we do. That the forest produces carbon, that it retains it, that it releases it, none of that makes no sense to them” (NGO worker, 2024).

Undergirding this conservation discourse, another well-known process is at play that can be best understood through the “environmentality” conceptual framework (Agrawal, 2005). Openly building on Foucault’s governmentality

paradigm, this conceptual framework has at its core the constitution of environmental subjects, “those for whom the environment constitutes a critical domain of thought and action” (Agrawal, 2005: 16). Like Foucault’s notion of biopolitics, it is attentive to the ways in which diverse technologies of government are applied to secure what he called the “conduct of conduct” (self-regulation), in this case within the environmental sphere of life and thought.

In a telling statement, a flower picker from Minas Gerais (Brazil), quoted by Haesbaert and Mason-Deese, states: “Invisibility guaranteed our way of life. Self-recognition came from the need to defend our territory” (2020: 261). For many Indigenous communities their self-construction as “mitigation actors” comes from the need to defend their territory. They have realized that their traditional lifestyle (or rather, a selective representation of it) aligns with the current global socio-technical arrangements and acts as a purported solution to a problem perceived by the global hegemonic system, i.e. periodically recurring limits to capital expansion (Harvey, 2006). In this sense, environmental “self-subjection,” consisting in the self-portrayal of all Indigenous people as infallible forest stewards, is instrumental to self-determination and resource sovereignty. Visibility, as Brighenti points out, can be “the flip side of control” (2007: 326) and surely there is no guarantee that “selling nature to save it” (McAfee, 1999) through carbon markets will be beneficial in the long run. Yet to many rural peoples the need to protect their territory is the utmost priority.

In this regard, it is important to keep in mind that the skillful use of the political discourse around nature and sustainability tends to be restricted to circles of Indigenous elites and representatives. Not all Indigenous groups are the same; to some, REDD-like forest conservation schemes—barring the neoliberal conception of nature that underpins them—might entail acceptable compromises in terms of forest uses. Yet “for some forest-dependent peoples, shifting cultivation is not so much ‘what they do’, as ‘who they are’” (Leggett and Lovell, 2012: 7), which introduces an identity dimension to the impact of conservation policies. The same applies to the Nükak; their nomadic lifestyle is not merely a practice of material reproduction, but an intimate identitarian dimension, as demonstrated—among other things—by their language. For instance, the word *chañüat* (“to live”) derives from *cha* (“to be”) and *ñü* (to walk in the forest). In other words, for the Nükak, to live is to be walking in the forest.

The Faustian bargain for some Indigenous groups lies in the prospect that, while trying to defend their land and customary lifestyle by making themselves visible by urbanized actors, they might in fact start to lose them. In Colombia, among urban dwellers and campesinos, it is commonplace to think that Indigenous people, including formerly nomadic groups, no longer want to live in the forest as they are now “civilized.” According to this logic, it makes no sense for Indigenous communities to hold inalienable rights on vast swathes of land if they are not using it for their ancestral practices nor employing it in ways that are deemed productive according to capitalist standards. This points to a crucial issue that goes beyond the potential social impact of carbon offsetting projects and dovetails with the original rationale of the emissions mitigation regime: Indigenous peoples have been effective forest stewards when forests had a value, not a price. A socioecological configuration in which forests on Indigenous land have a “price tag” on them is uncharted land.

One last—though not less concerning—impact of nature-based offsetting concerns its gender impact. To demonstrate additionality in REDD+ projects, carbon developers in Colombia have been known to impose a considerable reduction in the extension of *chagras*, an Indigenous agroforestry system (Díaz and Ruiz-Nieto, 2023). Since agricultural labor is mostly entrusted to women (men being typically more involved in hunting and fishing), *chagras* represent a space of political discussion and enunciation for Indigenous women. By taking part of this space away, REDD+ projects eat into women's political presence, thereby contributing to their marginalization and disempowerment. This runs counter to the stated goals of REDD+, especially those schemes that commit to CCB (Climate, Community and Biodiversity) and SD VIsta (Sustainable Development Verified Impact) Standards, which ostensibly contribute to the UN Sustainable Development Goals (Verra, n.d.).

Ultimately, then, besides the more obvious livelihood impact of reduced agricultural production, carbon projects come with a deeply political charge. When this is combined with the fact that Indigenous livelihood activities are often not the real driver of forest degradation or deforestation, carbon offsetting emerges as a particularly Machiavellian method to influence the resource sovereignty and social reproduction of Indigenous groups.

CONCLUSIONS

This paper has addressed nature-based carbon offsetting schemes and their potential direct and indirect impacts, with a particular focus on Colombia. While it can be argued that subtle green grabbing is directly caused by international actors operating on global carbon markets, the Colombian reality suggests more nuanced conclusions. Indeed, it appears that, at least for the time being, phenomena resembling green grabbing occur on a smaller, national scale “lens.” In Colombia this takes place on the back of a national law providing for the opportunity to legally bring down carbon tax bills by acquiring offsets with no requirement for international monitoring and a merely formal alignment to international standards (Dufrasne, 2021).

In this regard, however, it would be wrong to infer that these processes are totally independent from the global climate governance paradigm. In fact, it is that very paradigm that has construed carbon as a resource in the last few decades. Global links may not be immediately visible in the form of a neat commodity chain, yet a carbon forestry industry at any scale is imaginable only within a global context that accepts offsetting as a viable option to comply with climate commitments. Within this landscape, Gustavo Petro's attention to environmental issues ironically catalyzes this industry, whose real climate contribution is debatable. In this perspective, internal colonialism is enabled by a global green colonialism configuration (Lang, Manahan, and Bringel, 2024), in which Colombia is still entangled in a dependency network as a provider of poorly valued, “unprocessed” raw materials (the ability to preserve ecosystems) that are then turned into semi-processed goods (carbon credits purportedly abating the emissions caused by fossil fuel extraction and other activities; see Bermúdez Liévano, 2018), and finally fed into the production of processed goods worldwide.

Ultimately, the study of nature-based offsetting offers clear insights as to “[h]ow globalized green grabbing possibilities become localized” (Fairhead, Leach, and Scoones, 2012: 247). It reveals how offsetting, besides being a (relatively) new form of appropriation, is constituted not only by virtual markets but also by eminently material manifestations, involving “old” actors (authorities at various scales), new actors (NGOs, standardization bodies, certifiers, consultants, etc.), and more or less willing Indigenous communities. The latter are somehow left “at the doorstep” of the “complex encounters between science, technology and politics” (Fairhead, Leach, and Scoones, 2012: 241) that produce the carbon commodity.

To expect an outright rejection of carbon offsetting by these communities, however, may be a short-sighted assumption. As REDD+ often implies (or at least promises) an inflow of cash and a decentralization of forest governance, many in the Global South are willing to cross that doorstep (see for example FSC-IF, 2023). A representative of OPIAC (Organization of Indigenous Peoples of the Colombian Amazon) affirmed:

We are going to start with an information and training plan on REDD and much beyond REDD. . . . We are promoting a program that we have called the Colombian Amazon Indigenous School. We are not going to do this REDD+ Colombian Amazon Indigenous School on our own. We are going to have to partner with a university, we are going to have to partner with one, two, three NGOs, the WWF . . . and offer training at different levels. . . . Carbon markets are an opportunity for Indigenous peoples to create a different development model for the Amazon, but there must be enabling conditions, The government should create a public policy on issues beyond carbon, that is, let us call it an economy of nature (OPIAC member, 2024).

Carbon forestry—portrayed as a win-win climate solution and wielded by Indigenous peoples as a means for economic emancipation and sovereignty—emerges in this picture as the ultimate instrument to temper potential resistance to what could be called “land grabbing by carbon grabbing.” Going against it is to go against the undeferrable imperatives of biodiversity conservation and climate change mitigation, as well as against the empowerment of impoverished forest communities. Within the “there is no alternative”¹⁰ framework, the most likely scenario, especially in light of the continued lack of internationally agreed upon regulation,¹¹ is one in which governments and local authorities try to reign in market forces for the benefit of their constituents (this has been signaled, among others, by the Colombian Environment ministry - Marandua Stereo 100.7FM, 2023).

In this scenario, the need to co-design carbon offsetting projects with all stakeholders involved looms large. In countries like Colombia, where land tenure is not as much an issue as in other countries (most Indigenous communities own collective property over a *resguardo*), this process should ideally include not only formal landowners, but also all those actors that have formal or informal claims and influence over the territory. As complex as this may be, however, it comes up conspicuously short of ensuring historical and environmental justice. Since the carbon commodification process requires specific institutional scaffolding, technical documents, and specialist expertise (forms of technology, broadly speaking), a case could be made for technology transfer to rural

communities, which would make external developers redundant, thereby favoring an emancipatory use of carbon forestry.

Academia, for its part, has a role to play in continuously interrogating the structural problems with the commodification of nature and illuminating potential alternatives. If doing away with carbon forestry in the “real world” is not an option, then the goal of critical research should be to shed light on bad practices and power asymmetries, starting with those denounced by scholars and activists from the Global South. It remains an essential, if daunting, task to make sense of the ways in which climate capital “lands” in particular rural contexts, of what it brings with it and what it takes away, as well as of the local and global actors that act as intermediaries at the points of friction of global capitalism (Tsing, 2005).

NOTES

1. Relevant cases include the DRC (Nhantumbo and Samndong, 2013), Indonesia (Jong, 2023), Gabon (L, 2022), and the Ivory Coast (DGB, 2023).

2. While the extent to which pre-Columbian societies significantly altered forest environments not located along rivers is debated (Piperno et al., 2021), research has proven that their marks on tropical forests can still be detected today (Levis et al., 2017), particularly around archaeological sites. Such sites are likely to increase as LIDAR-based explorations reveal more earthworks beneath thick forest canopy (Peripato et al., 2023).

3. On this, see for example Fals Borda (1998).

4. Bolivia has long opposed carbon markets as a form of neocolonialism (see for example Ministerio de Relaciones Exteriores, 2021, and Gibson, 2023). In recent years, however, this opposition has dwindled. Bolivia is now developing the largest carbon forestry project in the world (GCFTE, 2023), covering 9 percent of the country’s forested area. Examples of attempts to regulate carbon markets include Brazil (see Duchelle et al., 2014; and ART, 2023) and Guyana (ART, 2022).

5. In the climate mitigation jargon, an additional activity is one that brings about emissions reductions that would not have taken place in the absence of that activity. In the case of REDD+, the (alleged) additionality lies in the fact that, without the project, the corresponding area of forest would have been affected by deforestation.

6. Estimates vary, see CEPAL (2013) and Murad and Pearse (2018).

7. For a historical overview and problematization of these concepts in Western academic literature, see for example Elden (2010); on landscape, see Wylie (2007).

8. Since this paper was written, the Constitutional Court has issued a somewhat ambiguous ruling that emphasises the centrality of informed consent by Indigenous peoples (see Bacca, 2024).

9. “Simply,” here, may not be the most accurate adverb as, according to Qin et al. (2021), carbon loss from forest degradation may exceed that from deforestation.

10. The reference here is to Margaret Thatcher’s well-known “There is no alternative” slogan.

11. At COP28 in Dubai, parties failed to reach an agreement on how to operationalize key carbon market regulations, namely articles 6.2 and 6.4 of the Paris Agreement (Zwick, 2023).

REFERENCES

- Agrawal, Arun
2005 *Environmentalism: Technologies of Government and the Making of Subjects*. Durham: Duke University Press.
- Amazon Carbon
n.d. “Deiyabena REDD Nükak Baka.” <https://www.amazoncarbon.bio/deiyabena-redd-nukak-baka/>.
- Amin, Samir
1976 *Unequal Development*. New York: Monthly Review Press.

ART

2022 "ART Issues World's First Jurisdictional Forestry TREES Carbon Credits to Guyana." <https://www.artredd.org/wp-content/uploads/2022/12/ART-Issues-Worlds-First-Jurisdictional-Forestry-TREES-Carbon-Credits-to-Guyana.pdf>.

2023 "ART Publishes TREES Concept for Mato Grosso, Brazil on the ART Registry." <https://www.artredd.org/art-publishes-trees-concept-for-mato-grosso-brazil-on-the-art-registry/>.

Asia Indigenous Peoples Pact

2010 "REDD+ Implementation in Asia and the concerns of Indigenous Peoples." http://www.iwgia.org/iwgia_files_publications_files/0654_REDD_Plus_Implementation_in_Asia_and_the_Concerns_of_Indigenous_Peoples.pdf.

Asiyanbi, Adeniyi and Jens Lund

2020 "Policy persistence: REDD+ between stabilization and contestation." *Journal of Political Ecology* 27 (1).

Asocarboneo

2023 "Informe actual sobre el estado actual del mercado colombiano de carbono a 30 Junio 2023." <https://asocarboneo.org/boletin-informativo-no-12/>.

Bacca, Paulo

2024 "Tenemos una sentencia sobre bonos de carbono. ¿Cómo debe ser la reglamentación?" *Dejusticia*. July 11, 2024. <https://www.dejusticia.org/column/tenemos-una-sentencia-sobre-bonos-de-carbono-como-debe-ser-la-reglamentacion/>.

Bermúdez Liévano, Andrés

2018. "El mayor proyecto de bonos de carbono de Colombia podría estar vendiendo aire caliente." *CLIP*. November 25, 2018. <https://www.elclip.org/el-mayor-proyecto-de-bonos-de-carbono-de-colombia-podria-estar-vendiendo-aire-caliente/?lang=es>.

2022 "El territorio de los jaguares de Yuruparí dividido por un proyecto de bonos de carbono." *CLIP*. October 26, 2022. <https://www.elclip.org/bonos-carbono-yurupari/?lang=es>.

Brighenti, Andrea

2007 "Visibility: A category for the social sciences." *Current Sociology* 55 (3): 323–342.

Bridge, Gavin

2009 "Material worlds: Natural resources, resource geography and the material economy." *Geography Compass* 3 (3): 1217–44.

Bringel, Breno and Maristella Svampa

2023 "Del 'Consenso de los Commodities' al 'Consenso de la Descarbonización.'" *Nueva Sociedad*. July–August, 2023. <https://nuso.org/articulo/306-del-consenso-de-los-commodities-al-consenso-de-la-descarbonizacion/>.

Bruna, Natacha

2022 "A climate-smart world and the rise of green extractivism." *Journal of Peasant Studies* 49 (4): 839–64.

Bumpus, Adam G. and Diana M. Liverman

2008 "Accumulation by decarbonization and the governance of carbon offsets." *Economic Geography* 84 (2): 127–55.

Büscher, Bram

2014 "Selling success: Constructing value in conservation and development." *World Development* 57: 79–90.

Casanova, Pablo

1965 "Internal colonialism and national development." *Studies in Comparative International Development* 1 (4): 27–37.

Castree, Noel

2003 "Commodifying what nature?" *Progress in Human Geography* 27 3: 273–297.

Cavanagh, Connor and Tor A. Benjaminsen

2014 "Virtual nature, violent accumulation: The "spectacular failure" of carbon offsetting at a Ugandan National Park." *Geoforum* 56 (September): 55–65.

Cardona, Diego

2014 *Trampas de REDD y de otros proyectos de conservación de bosques*. CENSAT Agua Viva - Amigos de la Tierra Colombia. <https://www.foei.org/wp-content/uploads/2020/12/Trampas-de-REDD-y-de-otros-proyectos-de-conservacionpercentCCpercent81n-de-bosques-1.pdf>.

CENSAT

2025 *Geo-grafiar. Mercados de carbono en Colombia*. CENSAT Agua Viva - Amigos de la Tierra Colombia. <https://geo-grafiarmc.com/>.

CEPAL

2013 *Amazonia posible y sostenible*. CEPAL. https://www.cepal.org/sites/default/files/news/files/folleto_amazonia_posible_y_sostenible.pdf.

Dávalos, Liliana M., Adriana C. Bejarano, Mark A. Hall, H. Leonardo Correa, Angelique Corthals, and Oscar J. Espejo

2011 "Forests and drugs: Coca-driven deforestation in tropical biodiversity hotspots." *Environmental Science & Technology* 45 (4): 1219–1227.

DGB

2023 "The Republic of Côte d'Ivoire partners with DGB Group on large-scale forest restoration projects." March 6. <https://www.green.earth/press-releases/ivory-coast-partners-dgb-forest-restoration>.

Díaz, Juan Manuel and Omar Ruiz-Nieto

2023. *Diagnóstico de proyectos REDD+ en la Amazonia colombiana*. Instituto Amazónico de Investigaciones Científicas Sinchi. <https://www.sinchi.org.co/diagnostico-de-proyectos-redd-en-la-amazonia-colombiana>.

Dove, Michael R.

1983 "Theories of swidden agriculture, and the political economy of ignorance." *Agroforestry Systems* 1: 85–99.

Duchelle, Amy E., Maron Greenleaf, Denyse Mello, Maria Fernanda Gebara, and Tadeu Melo

2014 "Acre's State System of Incentives for Environmental Services (SISA), Brazil," in Erin O. Sills, Stibniati S. Atmadja, Claudio de Sassi, Amy E. Duchelle, Demetrius L. Kweka, Ida Aju Pradnja Resosudarmo, and William D. Sunderlin (eds.), *REDD+ on the ground*. Bogor, Indonesia: CIFOR.

Dufresne, Gilles

2021 "Two Shades of Green: How hot air forest credits are being used to avoid carbon taxes in Colombia." *Carbon Market Watch*. June 30. <https://carbonmarketwatch.org/publications/two-shades-of-green-how-hot-air-forest-credits-are-being-used-to-avoid-carbon-taxes-in-colombia/>.

2023 "Secretive Intermediaries. Are carbon markets really financing climate action?" *Carbon Market Watch*. February 2. <https://carbonmarketwatch.org/publications/secret-intermediaries-are-carbon-markets-really-financing-climate-action/>.

Elden, Stuart

2010 "Land, terrain, territory." *Progress in Human Geography* 34 (6): 799–817.

Fals Borda, Orlando

1998 *Guía práctica del ordenamiento territorial en Colombia: Contribución para la solución de conflictos*. Instituto de Estudios Políticos y Relaciones Internacionales (IEPRI), Universidad Nacional de Colombia, Bogotá.

Fairhead, James, Melissa Leach, and Ian Scoones

2012 "Green grabbing: A new appropriation of nature?" *Journal of Peasant Studies* 39 (2): 237–61.

Franky, Carlos

2011 "Acompañarnos contentos con la familia." PhD diss., Wageningen University.

FSC-IF

2023 "Open Letter: Global South Voices in Support of REDD+." FSC-IF. <https://www.fscindigenousfoundation.org/global-south-voices-in-support-of-redd/>.

Garavito, Giovanni, Rafael Clavijo, Pilar Luengas, Pablo Palacios, and María Helena Arias

2021 "Assessment of biodiversity goods for the sustainable development of the chagra in an Indigenous community of the Colombian Amazon: Local values of crops." *Journal of Ethnobiology and Ethnomedicine* 17 (1): 23.

Garzón, Camilo, Manuela Galvis, and Antonia Diaz

2023 "Así es el mercado de bonos de carbono que el Estado, por ahora, no ve." *La Silla Vacía*. October 3. <https://www.lasillavacia.com/silla-nacional/asi-es-el-mercado-de-bonos-de-carbono-que-el-estado-por-ahora-no-ve/>.

Garzón, Camilo, Daniela Garzón and Antonia Diaz

2023 "La chagra amazónica enseña cómo producir comida sin acabar la selva." *La Silla Vacía*. October 5. <https://www.lasillavacia.com/silla-academica/la-chagra-amazonica-ensena-como-producir-comida-sin-acabar-la-selva/>.

GCFTF

2023 *Bolivia's Bold Step: Unveiling the World's Largest REDD+ Project for Sustainable Forest Management*. Governors' Climate and Forests Task Force. <https://www.gcftf.org/bolivia-jredd/>.

Gibson, Gibson

2023. "COP28: Bolivia proposes moratorium for Article 6.4 carbon markets, observers suggest "performative" move." *Carbon Pulse*. December 8. <https://carbon-pulse.com/243412/>.

Global Forest Watch (n.d.) Colombia. Guaviare. <https://www.globalforestwatch.org/dashboards/country/COL/16/>.

Gutiérrez, María

2011 "Making markets out of thin air: A case of capital involution." *Antipode* 43 (3): 639–61.

Gutiérrez, Laura, Nelsa De la Hoz, Natalia Hernández, and Diego Silva

2023 "The Colombian Amazon jungle: From house of the spirits to carbon sink." Conference presentation, ISEE 2023 Conference, Santa Marta, Colombia. October 26.

Haesbaert, Rogério and Liz Mason-Deese

2020. "Territory/ies from a Latin American perspective." *Journal of Latin American Geography* 19 (1): 258–68.

Harvey, David

2006 *The Limits to Capital*. London: Verso.

Haya, Barbara K., Kelsey Alford-Jones, William Anderegg, Betsy Beymer-Farris, Libby Blanchard, Barbara Bomfim, Dylan Chin, Samuel Evans, Marie Hogan, Jennifer A. Holm, Kathleen McAfee, Ivy So, Thales West, and Lauren Withey

2023 "Quality assessment of REDD+ carbon credit projects." *Berkeley Carbon Trading Project*. <https://gspp.berkeley.edu/research-and-impact/centers/cepp/projects/berkeley-carbon-trading-project/REDD+>.

Ingalls, Micah L. and Michael B. Dwyer

2016. "Missing the forest for the trees? Navigating the trade-offs between mitigation and adaptation under REDD." *Climatic Change* 136: 353–366.

Jong, Hans N.

2023 "Indonesia opens carbon trading market to both skepticism and hope." *Mongabay*. October 12. <https://news.mongabay.com/2023/10/indonesia-opens-carbon-trading-market-to-both-skepticism-and-hope/>.

Kavanagh, Michael J.

2023 "Billionaire Mark Cuban Backing \$1 Billion Congo Carbon Plan." *Bloomberg*. December 12. <https://www.bloomberg.com/news/articles/2023-12-12/billionaire-mark-cuban-backs-1-billion-congo-carbon-venture>.

L, Jennifer

2022 "Gabon Aims to Issue the Largest Volume of Carbon Credits Ever (187 Million)." *Carbon Credits*. July 1. <https://carboncredits.com/gabon-carbon-credits-biggest-issuance/>.

Lang, Miriam, Mary Ann Manahan and Breno Bringel

2024 *The Geopolitics of Green Colonialism. Global Justice and Ecosocial Transitions*. London: Pluto Press.

Leggett, Matthew and Heather Lovell

2011 "Community perceptions of REDD+: A case study from Papua New Guinea." *Climate Policy* 12 (1): 115–34.

Levis, C., F. R. C. Costa, F. Bongers, M. Peña-Claros, C. R. Clement, A. B. Junqueira, and E. G. Neves, et al.

2017 "Persistent Effects of Pre-Columbian Plant Domestication on Amazonian Forest Composition." *Science* 355 (6328): 925–31.

Leonardi, Emanuele

2017 "Carbon trading dogma: Theoretical assumptions and practical implications of global carbon markets." *Ephemera: Theory & Politics in Organization* 17 (1): 61–87.

Mahecha, Dany and Carlos Franky

2013 "The Nükak: The Last Nomadic People Officially Contacted in Colombia," in Dinah Shelton, Antenor Vaz, Beatriz Huertas Castillo, Carlos Camacho Nassar, Luis Jesús Bello, Paola Colleoni, José Proaño, Dany Mahecha, Carlos Franky, Unión de Nativos Ayoreo de Paraguay, and Iniciativa Amotocodie (eds.), *Indigenous Peoples In Voluntary Isolation And Initial Contact*. IWGIA.

Marandua Stereo 100.7FM

2023 "Ministra de Ambiente, Susana Muhamad en #MaranduaVerde." *Facebook*. February 25. Video. <https://www.facebook.com/maranduastereo/videos/1246550932936337/>.

Marawanyika, Godfrey and Antony Sguazzin

2023 "Dubai Firm Wants a Fifth of Zimbabwe Landmass for Carbon Credits." *Bloomberg*. September 29. <https://www.bloomberg.com/news/articles/2023-09-29/zimbabwe-uae-firm-sign-1-5-billion-carbon-credit-financing-mou#xj4y7vzkg>.

Mayorga Alejo, Carlos

2023 "Con aplausómetro, Biofix logró consentimiento indígena para negociar bonos de carbono." *Verdadabierta.com*. July 23. <https://verdadabierta.com/con-aplausometro-biofix-logro-consentimiento-indigena-para-negociar-bonos-de-carbono/>.

McAfee, Kathleen

1999 "Selling nature to save it? Biodiversity and green developmentalism." *Environment and Planning D: Society and Space* 17 (2): 133–154.

Mendoza, Diana

2024 "El Mundo Indígena 2024: Colombia." IWGIA. April 12. https://iwgia.org/es/colombia/5481-mi-2024-colombia.html#_edn4.

Ministerio de Ambiente y Desarrollo Sostenible

2024 *Monitoreo de la superficie de bosque y la deforestación en Colombia – 2023*. IDEAM. <https://drive.google.com/drive/folders/1G7S8eXtZ2O2xHOiUO7JNlQKJ4j8i3duK>.

Ministerio de Relaciones Exteriores

2021 *Bolivia alerta sobre el "nuevo colonialismo del carbono" en la COP 26*. Ministerio de Relaciones Exteriores. <https://www.cancilleria.gob.bo/webmre/node/4681>.

Mongabay

2022 "Indígenas negocian bonos de carbono en desventaja y sin respaldo estatal." *Mongabay*. September 1. <https://es.mongabay.com/2022/09/indigenas-negocian-bonos-de-carbono-en-desventaja-y-sin-respaldo-estatal-en-colombia/>.

Moros, Lina, Esteve Corbera, María Alejandra Vélez, and Daniel Flechas

2020 "Pragmatic conservation: Discourses of payments for ecosystem services in Colombia." *Geoforum* 108: 169–183.

MSCI

2023 "Voluntary Carbon Market in Review." Trove Research. PowerPoint presentation.

Murad, Cesar A. and Jillian Pearse

2018 "Landsat study of deforestation in the Amazon region of Colombia: Departments of Caquetá and Putumayo." *Remote Sensing Applications: Society and Environment* 11: 161–171.

Nhantumbo, Isilda and Raymond Samndong

2023 "REDD+ and rights: Extending carbon rights in the DRC to climate-regulating services." IIED Briefing. <https://www.iied.org/sites/default/files/pdfs/migrate/17182IIED.pdf>.

ONIC

n.d "Pueblos indígenas." <https://www.onic.org.co/noticias/2-sin-categoria/1038-pueblos-indigenas>.

Pérez Lara, Martín, Luz Gómez Gallego, and Julián Andrade Franco

2023 "Comportamiento de la deforestación en proyectos REDD+ en Colombia." CCAP and FCDS, 2023. <https://fcds.org.co/wp-content/uploads/2023/02/vf-document-deforestation-in-redd-projects-in-colombia.pdf>.

Peripato, Vinicius, Carolina Levis, Guido A. Moreira, Dani Gamerman, Hans ter Steege, Nigel C.A. Pitman, Jonas G. de Souza, José Iriarte, and Mark Robinson, et al.

2023 "More than 10,000 pre-Columbian earthworks are still hidden throughout Amazonia." *Science* 382 (6666): 103–109.

Phelps, Jacob, Edward L. Webb, and Arun Agrawal

2010 "Does REDD+ threaten to recentralize forest governance?" *Science* 328 (5976): 312–13.

Piperno, Dolores R., Crystal H. McMichael, Nigel C. A. Pitman, Juan Ernesto Guevara Andino, Marcos Ríos Paredes, Britte M. Heijink, and Luis A. Torres-Montenegro

2021 "A 5,000-Year vegetation and fire history for *Tierra Firme* forests in the Medio Putumayo-Algodón watersheds, Northeastern Peru." *Proceedings of the National Academy of Sciences* 118 (40): e2022213118.

Qin, Yuanwei, Xiangming Xiao, Jean-Pierre Wigneron, Philippe Ciais, Martin Brandt, Lei Fan, and Xiaojun Li, et al.

2021 "Carbon Loss from Forest Degradation Exceeds That from Deforestation in the Brazilian Amazon." *Nature Climate Change* 11 (5): 442–48.

Renare

2025 "Listado de iniciativas." IDEAM. <https://renare.ideam.gov.co/GPY2-web/#/gpy/iniciativas>.

Rizzi, Andrea

2025 "Ir-REDD-ucible: Addressing the Policy Persistence of Nature-Based Offsetting." *ACME: An International Journal for Critical Geographies* 24 (1): 108–134.

Rodríguez-de-Francisco, Jean Carlo, Carlos Del Cairo, Daniel Ortiz-Gallego, Juan Sebastian Velez-Triana, Tomás Vergara-Gutiérrez, and Jonas Hein

2021 "Post-Conflict Transition and REDD+ in Colombia: Challenges to Reducing Deforestation in the Amazon." *Forest Policy and Economics* 127 (June): 102450.

Schroeder, Heike and Nidia C. González

2019 "Bridging knowledge divides: The case of Indigenous ontologies of territoriality and REDD+." *Forest Policy and Economics* 100 (March): 198–206.

Setyowati, Abidah

2020 "Governing the ungovernable: Contesting and reworking REDD+ in Indonesia." *Journal of Political Ecology* 27 (1).

Skutsch, Margaret and Esther Turnhout

2020. "REDD+: If communities are the solution, what is the problem?" *World Development* 130 (June): 104942.

Sponsel, Leslie E., Thomas N. Headland and Robert C. Bailey

1996 *Tropical Deforestation: The Human Dimension*. New York: Columbia University Press.

Thomson, Frances

2019 "The political economy of land and dispossession in Colombia." PhD diss., University of Sussex.

Tienhaara, Kyla

2012 "The potential perils of forest carbon contracts for developing countries: Cases from Africa." *Journal of Peasant Studies* 39 (2): 551–72.

Tsing, Anna

2005 *Friction: An Ethnography of Global Connection*. Princeton: Princeton University Press.

Ulloa, Astrid

2013 "Controlando la naturaleza: ambientalismo transnacional y negociaciones locales en torno al cambio climático en territorios indígenas en Colombia." *Iberoamericana* 13 (49): 117–133.

Velásquez Ruiz, Marco

2018 *Collective land tenure in Colombia*. CIFOR. https://www.cifor-icraf.org/publications/pdf_files/infobrief/6902-infobrief.pdf.

Verra

n.d. "Standards for a Sustainable Future." <https://verra.org/wp-content/uploads/2019/01/Verra-Standards-Factsheet.pdf>.

Wylie, John

2007 *Landscape*. Abingdon: Routledge.

Zwick, Steve

2023 "Countries Reject Proposed Guidance for Article 6." *Ecosystem Marketplace*. December 12. <https://www.ecosystemmarketplace.com/articles/cop-28-enters-final-day-with-no-end-in-sight/>.